



## Area Evaluation

**Question:** "Given that the actual geographical area of the entire image is 42025.0 square meters, what is the approximate tree-covered area of the ecosystem(Can be calculated based on coverage ratio)?

**Options:** A: [39700-39800]square meters B: [9500-9600]square meters C: [10300-10400]square meters D: [2600-2700]square meters

**Answer:** B



### Reasoning:

1. The question provides the total geographical area of the image as 42,025 square meters and asks for the approximate tree-covered area based on the coverage ratio.
2. Observing the image, tree-covered areas are concentrated along the riverbank, around houses, and in open spaces, but they do not form a majority.
3. The tree-covered area appears to occupy a moderate portion, estimated at around 22.6-22.8% of the total area.
4. Calculating this percentage of 42,025 square meters:  $0.227 * 42,025 \approx 9540$ . This value is close to the range [9500-9600] square meters.
5. Comparing this result with the options, Option B ([9500-9600] square meters) is the most appropriate choice.
6. Therefore, the answer is B.

**GPT-4.1:**  [bert\_f1: 0.59] [rougel\_f: 0.51]

1. The question provides the total geographical area of the image as 42,025.0 square meters and asks for the approximate tree-covered area based on the coverage ratio.
2. By visually analyzing the image, the tree-covered regions are mainly along the water, scattered throughout the residential lots, and along the roads, but they do not dominate the entire image. The majority of the area consists of open land, houses, roads, and water.
3. Estimating the tree coverage, it appears to be roughly 25% of the total area. Calculating 25% of 42,025 gives approximately 10,506 square meters. Among the options, the range [10,300-10,400] square meters (Option C) is the closest to this estimate.
4. Therefore, the correct answer is C. Answer: C

**InternVL3-38B:**  [bert\_f1: 0.16] [rougel\_f: 0.45]

1. The image shows a landscape with a significant portion covered by trees, particularly on the left side and around the houses.
2. The total area of the image is given as 42025.0 square meters.
3. By visually estimating the tree coverage, it appears that trees cover roughly a quarter to a third of the total area.
4. Calculating a quarter of 42025.0 square meters gives approximately 10506.25 square meters.
5. This value falls within the range of option C: [10300-10400] square meters.
6. Therefore, option C is the most appropriate choice based on the estimated tree coverage ratio. Answer: C